NAME: _____ Geometry (FL B.E.S.T.)

5.1 Dilations

A **dilation** is a transformation that scales a figure from a fixed **center** by a **scale factor** k, producing a similar image. For a dilation centered at the origin, the rule is $(x, y) \mapsto (kx, ky)$; for a center at (a, b), translate the center to the origin first: $(x, y) \mapsto (k(x-a)+a,\ k(y-b)+b)$. When $|k| > 1$ the image is an **enlargement**; when 0<|k|<1 it is a **reduction**; a negative k places the image on the opposite side of the center, along the same line through the center and the original point. Lengths scale by a factor of $|k|$, but **areas scale by** k^2 (since both dimensions grow). The center itself is the only fixed point — it always maps to itself. This worksheet practises dilating points and figures, finding scale factors, and reasoning about how length and area change.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Find the image of $A(2, 5)$ under a dilation about the origin with $k = 4$.

2. Find the image of $B(-3, 6)$ under a dilation about the origin with $k = \frac{1}{3}$.

3. Under a dilation about the origin, $P(8, 12)$ maps to $P'(2, 3)$. Find k.

4. Under a dilation about the origin, $Q(5, -4)$ maps to $Q'(15, -12)$. Find k.

5. Is a dilation with $k = 0.6$ an enlargement or a reduction?

6. A segment of length 10 is dilated with $k = 0.4$. Find the image length.

7. Dilate $S(2, 4)$ about center $P(2, 0)$ with $k = 2$.

8. $\triangle DEF$ has vertices $D(0, 0)$, $E(4, 0)$, $F(0, 6)$. Dilate about the origin with $k = \frac{1}{2}$.

9. A segment of length 14 is dilated with $k = \frac{3}{2}$. Find the image length.

10. Find the image of $C(7, -1)$ under a dilation about the origin with $k = -2$.

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11. Dilate $T(4, 6)$ about center $O(0, 2)$ with $k = \frac{1}{2}$.

12. Under a dilation about the origin, $M(-2, 8)$ maps to $M'(-5, 20)$. Find k.

13. A square has side length 6 (area 36). It is dilated with $k = \frac{2}{3}$. Find the new side length, the new area, and verify the ratio of areas equals k^2.

14. Dilate $W(6, 2)$ about center $C(2, 2)$ with $k = -3$. Find the image, then state where it lies relative to W and C.

15. A model car uses scale factor $k = \frac{1}{12}$ of the real car. The real car is 180 in long with a side-profile area of 2880 in^2. Find the model's length and side-profile area.

✓ Answer Key

1. $A'(8, 20)$
2. $B'(-1, 2)$
3. $k = \frac{1}{4}$
4. $k = 3$
5. Reduction
6. 4
7. $S'(2, 8)$
8. $D'(0, 0), E'(2, 0), F'(0, 3)$
9. 21
10. $C'(-14, 2)$
11. $T'(2, 4)$
12. $k = \frac{5}{2}$
13. New side $= 4$; new area $= 16$; ratio $= \frac{4}{9} = k^2$
14. $W'(-10, 2)$; on the opposite side of C from W, three times as far away
15. Model length $= 15$ in; model area $= 20$ in^2